Practical No. 7

Multi-Node Hadoop Cluster for Business Intelligence

Aim

To configure a multi-node Hadoop cluster and process large datasets for business intelligence analysis.

Learning Outcomes

- Understand Hadoop cluster architecture
- Configure multi-node HDFS
- Process big data using MapReduce

Software & Hardware Requirements

Hardware:
- Minimum 8 GB RAM

Software:
- Apache Hadoop
- Ubuntu Linux
- Java JDK

Dataset Link

Retailrocket Ecommerce Dataset – user events & product interactions (great for BI): https://www.kaggle.com/datasets/retailrocket/ecommerce-dataset (Kaggle)

Theory

A multi-node Hadoop cluster distributes storage and computation across multiple nodes, improving scalability and performance for business intelligence workloads.

System Architecture

Data Nodes → HDFS
↓
MapReduce Jobs
↓
BI Reports

Procedure

Step 1: Configure NameNode and DataNodes
Step 2: Start Hadoop cluster
Step 3: Upload dataset to HDFS
Step 4: Run MapReduce job

Output

- Processed BI reports
- HDFS data distribution

Result

A multi-node Hadoop cluster was successfully configured and executed.

Applications

- Enterprise analytics
- Data warehousing
- Business intelligence systems

Conclusion

This experiment highlights Hadoop’s scalability for BI solutions